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APPLIED MICROECONOMETRICS

Group Assignment

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1 Introduction

2 Model and estimators

2.1 Static panel data model

We consider a static panel data model with both time-varying and time-invariant regressors. The model is given by

$$y_{it} = \beta_u x_{u,it} + \beta_c x_{c,it} + \gamma_u z_{u,i} + \gamma_c z_{c,i} + c_i + u_{it},$$

where $i = 1, \dots, N$ indexes individuals and $t = 1, \dots, T$ indexes time. The variable c_i denotes an unobserved individual-specific effect, while u_{it} is an idiosyncratic error term. The regressors are divided into four groups, as shown in Table 1. Here, exogenous means uncorrelated with the individual effect c_i , while endogenous means correlated with c_i .

This distinction is important because the correlation between regressors and unobserved individual effects determines which panel data estimators are appropriate. If regressors are correlated with c_i , estimators that treat c_i as random and independent of the regressors will generally be biased. Time-invariant regressors create an additional challenge in panel data analysis because some commonly used methods rely exclusively on within-individual variation over time. Since time-invariant variables do not vary across periods for a given individual, their effects may not be identifiable under such transformations.

Table 1: Classification of the regressors

Variable	Time dimension	Relation with c_i
x_u	time-varying	exogenous
x_c	time-varying	endogenous
z_u	time-invariant	exogenous
z_c	time-invariant	endogenous

2.2 Estimators

2.2.1 Pooled ordinary least squares

As a first benchmark, we estimate the model using pooled ordinary least squares (POLS). Let $w_{it} = (x_{u,it}, x_{c,it}, z_{u,i}, z_{c,i})'$ denote the vector of regressors. Pooled OLS estimates the parameter vector by minimizing the sum of squared residuals over all observations, treating the model as a standard cross-sectional regression. In doing so, it ignores the panel structure and absorbs the individual effect c_i into the composite error term $v_{it} = c_i + u_{it}$. Consistency therefore requires $E(w_{it}v_{it}) = 0$. Since $x_{c,it}$ and $z_{c,i}$ are correlated with c_i by construction, this condition is violated, implying that pooled OLS is generally biased and inconsistent. Nevertheless, it provides a useful baseline against which the panel estimators can be compared.

2.2.2 Fixed effects

The second estimator is the fixed effects (FE) estimator. FE removes the individual-specific c_i by applying the within transformation, which subtracts the individual means from each variable:

$$y_{it} - \bar{y}_i = \beta_u(x_{u,it} - \bar{x}_{u,i}) + \beta_c(x_{c,it} - \bar{x}_{c,i}) + (u_{it} - \bar{u}_i).$$

The time-invariant variables $z_{u,i}$, $z_{c,i}$ and c_i disappeared because their values do not vary over time, so their individual means equal the variables themselves.

2.2.3 Random effects

The third estimator is the random effects (RE) estimator. Random effects exploits the error-components structure ($v_{it} = c_i + u_{it}$) and applies generalized least squares to account for the covariance induced by the individual effect. The estimator can be viewed as a quasi-demeaning transformation,

$$y_{it} - \theta \bar{y}_i = (w_{it} - \theta \bar{w}_i)' \delta + (u_{it} - \theta \bar{u}_i),$$

where θ depends on the variance components of c_i and u_{it} . Unlike fixed effects, RE retains variation in time-invariant regressors and therefore allows estimation of γ_u and γ_c . However, consistency requires the strict exogeneity condition $E(c_i | w_{it}) = 0$ for all t . This assumption is violated in our simulation design for $x_{c,it}$ and $z_{c,i}$, so the RE estimator is expected to be biased in the presence of endogenous regressors.

2.2.4 Correlated random effects

2.2.5 Hausman-Taylor

3 Simulation design

4 Performance measures

5 Results

6 Conclusion